

CyberLaw Bot

Improvements & Advancement Roadmap

Prepared for the Bano Qabil National Hackathon

Current State

CyberLaw Bot is a RAG-based AI legal assistant for Pakistani cyber and criminal law (PECA 2016, PECA 2025 Amendment, PPC, CrPC), built with FastAPI, the Gemini API, and a custom vector search implementation (no LangChain/LangGraph). It is deployed on Render at cyberlawbotpk.app with rate limiting, CORS restrictions, and environment-variable secret management.

Precision@6	88.7%
Relevance	90.2%
Citation accuracy	100%
Response time	Sub-5 seconds

1. Fix Before Demo Day

Visible issues that should be resolved before any judge opens the live site.

- **Fix the zoom-blocking viewport bug** — Remove `maximum-scale=1.0` and `user-scalable=no` from the viewport meta tag — this currently blocks pinch-to-zoom, a WCAG 1.4.4 accessibility failure that's especially visible on mobile/tablet.
- **Add a Privacy Policy / Terms of Service page** — For a legal-AI tool handling user queries, the absence of a privacy policy is a real gap that judges evaluating responsible-AI or social-impact criteria will notice.
- **Add an About / How-it-works section** — Add a one-screen explainer of what the tool does, the tech stack, and the accuracy metrics — right now the site is only the chat widget, so the impressive numbers behind it are invisible to a visitor.

2. Technical Differentiators to Build

Hackathons reward visible technical depth over invisible backend quality — these turn backend work into things judges can see and test.

- **Source-linking on citations** — When the bot cites a PECA/PPC/CrPC section, link or expand to the actual statute text inline — this visually proves the 100% citation accuracy claim in real time during a live demo.
- **Confidence / relevance indicator** — A small UI element showing retrieval confidence per answer reinforces that this is grounded retrieval, not an LLM guessing, and highlights the custom vector search work.
- **Multi-turn conversation memory** — If currently single-shot Q&A, add context retention so follow-up questions (e.g. "what about a repeat offense?") work naturally — judges will test this.
- **Latency display** — Show a small "Answered in 2.3s" tag under each response — turns the sub-5s backend metric into a visible, felt feature.

- **PECA 2016 vs. 2025 Amendment comparison** — A side-by-side "what changed" view — shows the dataset is actively maintained, not static, and is a distinct KP/Pakistan-specific legal-currency story.
- **RAG transparency panel** — Show the actual retrieved statute chunks used to generate an answer before the final response — proves the RAG pipeline is real, not a citation slapped onto a generic LLM answer.

3. Reach & Accessibility

- **WhatsApp integration** — WhatsApp reach in Pakistan dwarfs standalone web apps — a WhatsApp bot version is a strong 'impact at scale' story for judges.
- **Pashto language support** — Being KP-based, adding Pashto alongside Urdu/English is a concrete, defensible regional localization story.
- **Voice input/output** — Speech-to-text query plus text-to-speech answer (Urdu-first) helps users with lower literacy — a genuine accessibility differentiator, not a gimmick.
- **Low-bandwidth / PWA mode** — Offline caching of common Q&A or a lightweight PWA matters given inconsistent connectivity outside major cities.

4. Trust & Safety

- **User feedback loop** — Thumbs up/down per answer feeding an eval set — shows an iteration loop, not just a one-time build.
- **Severity / urgency flagging** — If a query suggests an active emergency (e.g. ongoing harassment or blackmail), surface relevant helpline numbers (e.g. FIA Cybercrime Wing) alongside the legal answer — a strong social-impact signal.
- **No PII retention** — Explicitly state and enforce that queries aren't stored with identifying information — pairs directly with the privacy policy addition above.

5. Viability — "What's Next" Story

- **Lawyer referral layer** — Route flagged complex cases to verified lawyers — turns 'legal info, not advice' into a sustainable model via referral fees, and answers the inevitable judge question about post-hackathon scaling.
- **API for other legal-tech builders** — Position the retrieval engine as infrastructure other Pakistani legal-tech builders could use, not just an end-user chatbot.
- **Local/smaller model roadmap** — Flag a roadmap toward a locally-hosted or fine-tuned model, since queries currently go through a foreign (Gemini) API — addresses data-sovereignty concerns for judges evaluating responsible-AI criteria.

6. Demo-Day Prep

- **Seed reliable demo queries** — Have 2-3 pre-tested prompts, including the Urdu one, that reliably showcase retrieval accuracy and bilingual support — avoid improvising live.
- **Have an offline fallback** — If Render cold-starts or venue WiFi is bad, a short screen-recording backup avoids a dead-air demo.
- **Polish rate-limit messaging** — Make sure the rate-limit response is a friendly message, not a raw 429, in case a judge or crowd hammers the bot during the demo.

Judging Framing

Pakistan's cyber law landscape (PECA 2016 + 2025 Amendment) is genuinely under-served by accessible legal-information tools. Stating this explicitly in the pitch — how many people currently have no easy way to check PECA penalties before something like a defamation complaint or FIR, and how this tool closes that gap in plain Urdu/English — directly addresses the social-impact and market-feasibility criteria hackathon judges typically weigh alongside technical execution.